

ARJUN VARMA

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EDUCATION

Columbia University

New York, NY

Master of Science in Data Science, 3.6/4.0

Dec 2026

Coursework: Applied Machine Learning, Agentic AI for Analytics, Statistical Inference and Modeling, Probability and Statistics

Teaching Assistant, Columbia Business School: Business Analytics II (Foundations of AI) and Hollywood and Big Data

Vellore Institute of Technology

Vellore, IN

Bachelor of Technology in Electronics and Communication Engineering, 4.0/4.0

May 2022

Special Achiever Award | Merit Scholarship

WORK EXPERIENCE

ZS Associates

Pune, IN

Advanced Data Science Associate Consultant

Feb 2025 - Jun 2025

- Built and deployed an org-wide analytics and ML platform (Spark/SQL, dashboards) that unified 5+ data sources into territory and product KPIs used by 100+ stakeholders supporting a \$10B oncology portfolio.
- Cut weekly reporting time from days to minutes, replacing Excel workflows with automated pipelines and self-serve dashboards

ZS Associates

Pune, IN

Decision Analytics Associate Consultant

Jul 2024 - Jan 2025

- Led a 5-member team to modernize legacy business rules; saved ~50 hrs/mo and improved first-pass quality to >99%
- Built and deployed Positive-Unlabeled learning models to infer missing categorical labels in medical transaction data, increasing customer-journey analytics coverage from ~40% to ~95% with consistent performance across tumor types and territories
- Implemented drift monitoring (feature + prediction drift) and CI unit tests for production pipelines, reducing silent failures
- Placed in the top ~10% in a company-wide hackathon and earned mselection for a lateral transfer into the Data Science vertical

ZS Associates

Pune, IN

Decision Analytics Associate

Feb 2022 - Jun 2024

- Engineered PySpark/SQL ETL pipelines integrating multiple healthcare data sources covering millions of patients for \$4B+ oncology drug performance analytics
- Defined patient cohort inclusion and exclusion logic robust to missing and miscoded fields, enabling audit-ready reporting
- Delivered ad hoc analyses identifying care gaps and market opportunities to inform brand strategy across multiple launches
- Promoted to Associate Consultant in 4 cycles (typical: 5) and received Expert Associate and Insight Illuminator awards

PROJECT EXPERIENCE

Biliary Tract Cancer (BTC) early detection using ML - Python, scikit-learn, PySpark

- Developed an early detection model across 250M patient claims, enabling ~45-day earlier identification compared standard diagnosis lag
- Engineered a hybrid feature pipeline combining clinical risk factors, K-means and GMM segmentation and Transformer-based NLP clustering on diagnosis narratives
- Presented methodology at [PMSA 2025](#) conference and the model was adopted for territory-level resource planning

Financial RAG Chatbot - Python, FastAPI, ChromaDB, Streamlit

[Github/Web Link](#)

- Built an LLM-powered RAG chatbot answering company financial questions from SEC filings with line-level citations
- Implemented semantic retrieval with ChromaDB and text-embedding-3-large plus automatic ticker and period parsing
- Built a multi-model evaluation pipeline using Claude Opus as a judge and deployed a live demo on Streamlit Cloud

Agricultural Product Standardization & Risk Detection - Sunculture (Series B Agtech)- Python, RAG, REST API

- Built a RAG-augmented classification system categorizing 7M+ farmer transactions across 500+ product categories to support creditworthiness assessment for microloans in East Africa
- Achieved 99% accuracy on a 10,000-item holdout set using hybrid rule-based and LLM-assisted classification, reducing manual review volume by 95% and accelerating loan decisioning

Multi-Agent Historical Dialogue System - Python, Flask, OpenRouter API

[Github/Web Link](#)

- Built AI chatbot enabling conversations with 60+ historical figures using multi-model LLM support and streaming responses
- Implemented era-appropriate prompt engineering and "Dinner Party" mode for multi-figure conversations; deployed on Railway

SKILLS

Programming: Python, SQL, R, C++

Analytics & ML: classification, regression, NLP, clustering, model evaluation, LLMs/RAG, prompt engineering

Data & MLOps: Spark (PySpark), Databricks, ETL/ELT, MLflow, Docker, AWS (S3, EMR, Athena, SageMaker)

Libraries: pandas, NumPy, scikit-learn, PyTorch, SHAP, matplotlib, BeautifulSoup

Tools: Git, Linux, CI/CD, Jupyter, Jira, Confluence, Claude Code, Cursor IDE